



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008 Big Data Tools and Technologies

What is a OLAP Cube



Online analytical processing (OLAP) cubes are used on the existing data warehouse infrastructure to provide self-service business intelligence capabilities to end users.

An OLAP cube is a data structure that overcomes the limitations of relational databases by providing rapid analysis of data. Cubes can display and sum large amounts of data while also providing users with searchable access to any data points. This way, the data can be rolled up, sliced, and diced as needed to handle the widest variety of questions that are relevant to a user's area of interest.

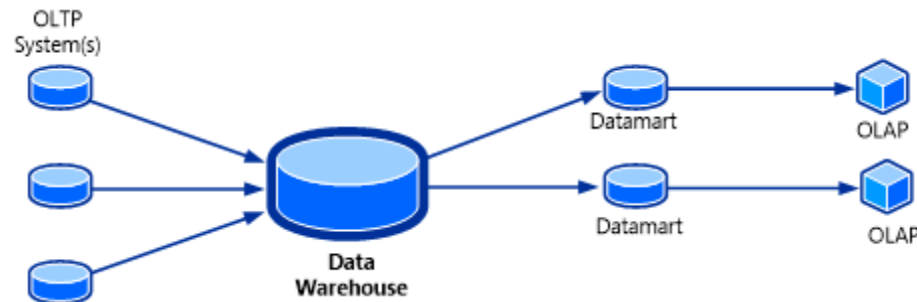
What is a OLAP Cube - Continued



Software vendors or information technology (IT) developers with a working knowledge of OLAP cubes can create management packs to define their own extensible and customizable OLAP cubes that are built on the data warehouse infrastructure. These cubes are stored in SQL Server Analysis Services (SSAS). Self-service business intelligence tools such as Excel and SQL Server Reporting Services (SSRS) or Power BI /PowerPivot can target these cubes in SSAS, and you can use them to analyze the data from multiple perspectives.

What is a OLAP Cube - Continued

OLAP cubes can be considered as the final piece of the puzzle for a data warehousing solution. An OLAP cube, also known as multidimensional cube or hypercube, is a data structure in SQL Server Analysis Services (SSAS) that is built, using OLAP databases, to allow near-instantaneous analysis of data. The topology of this system is shown in the following illustration.



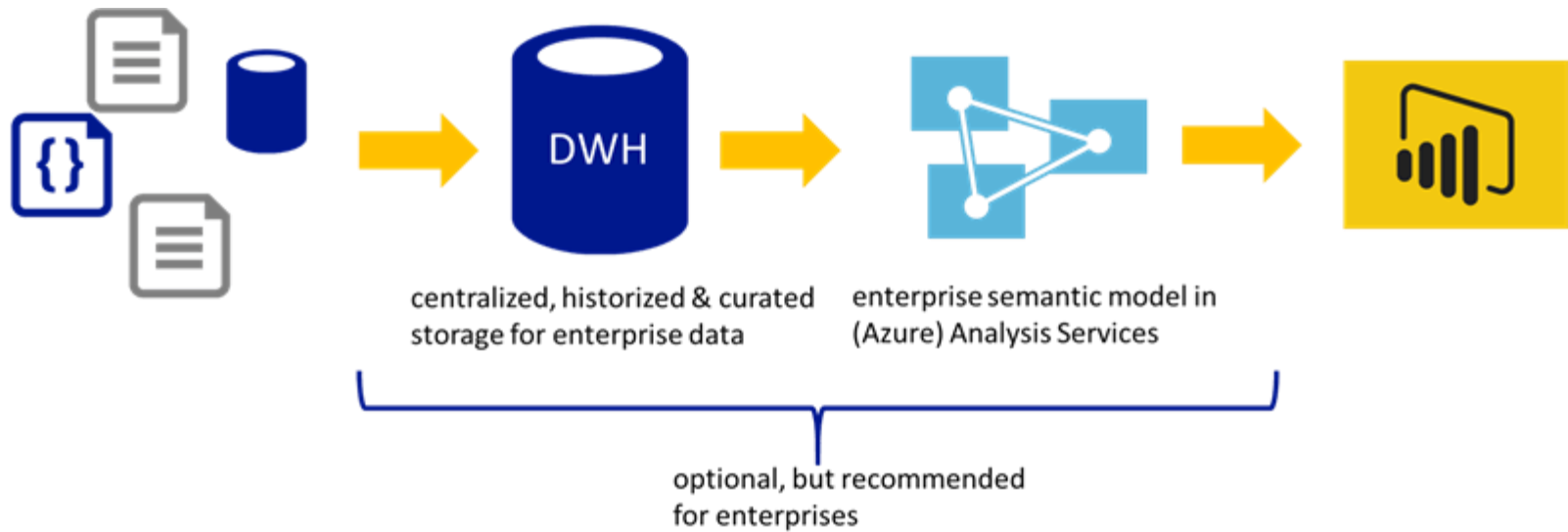
What is a OLAP Cube - Continued



The useful feature of an OLAP cube is that the data in the cube can be contained in an aggregated form. To the user, the cube seems to have the answers in advance because assortments of values are already precomputed. Without having to query the source OLAP database, the cube can return answers for a wide range of questions almost instantaneously.

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Reference architecture



Step 1 – New Project

Create a new project

Recent project templates

 Integration Services Project

analysis



Clear all

All languages

All platforms

All project types



Analysis Services Multidimensional and Data Mining Project

An Analysis Services project for creating multidimensional and data mining models.



Analysis Services Multidimensional and Data Mining Project

An Analysis Services project for creating multidimensional and data mining models.



Analysis Services Tabular Project

An Analysis Services project for creating tabular models.

New



Import from Server (Multidimensional and Data Mining)

Creates a multidimensional and data mining project by extracting the metadata from an existing multidimensional and data mining database on an Analysis Services server.

Step 1.5 – configure project



Configure your new project

Analysis Services Tabular Project

Project name

cubeprocessing_tabular_azureSSAS

Location

C:\Users\azureuser\Source\Repos

...

Solution

Create new solution

Solution name ⓘ

cubeprocessing_tabular_azureSSAS

☐

Place solution and project in the same directory

Framework

.NET Framework 4.7.2

Step 2 – Point to SSAS



Tabular model designer

Select an Analysis Services instance to use while authoring projects.

☐ Integrated workspace

Integrated workspace eliminates the need to provide an explicit AS server instance.

☒ Workspace server

asazure://eastus.asazure.windows.net/nizaralitabulardmodel

Test Connection

In order to create a new Tabular model, you must select a compatibility level. The compatibility level must be compatible with the Analysis Services server version you want to deploy to.

Compatibility level:

SQL Server 2016 RTM (1200)

[Click here for more information about compatibility level.](#)

☐ Do not show this message again.

OK Cancel

Step 3 – Impersonation

Table Import Wizard

Impersonation Information
Specify the credentials used by the Analysis Services server to connect to the data source when importing and processing data.

☐ Specific Windows user name and password
Connects to the data source using the credentials of the user named below.

User Name:
Password:

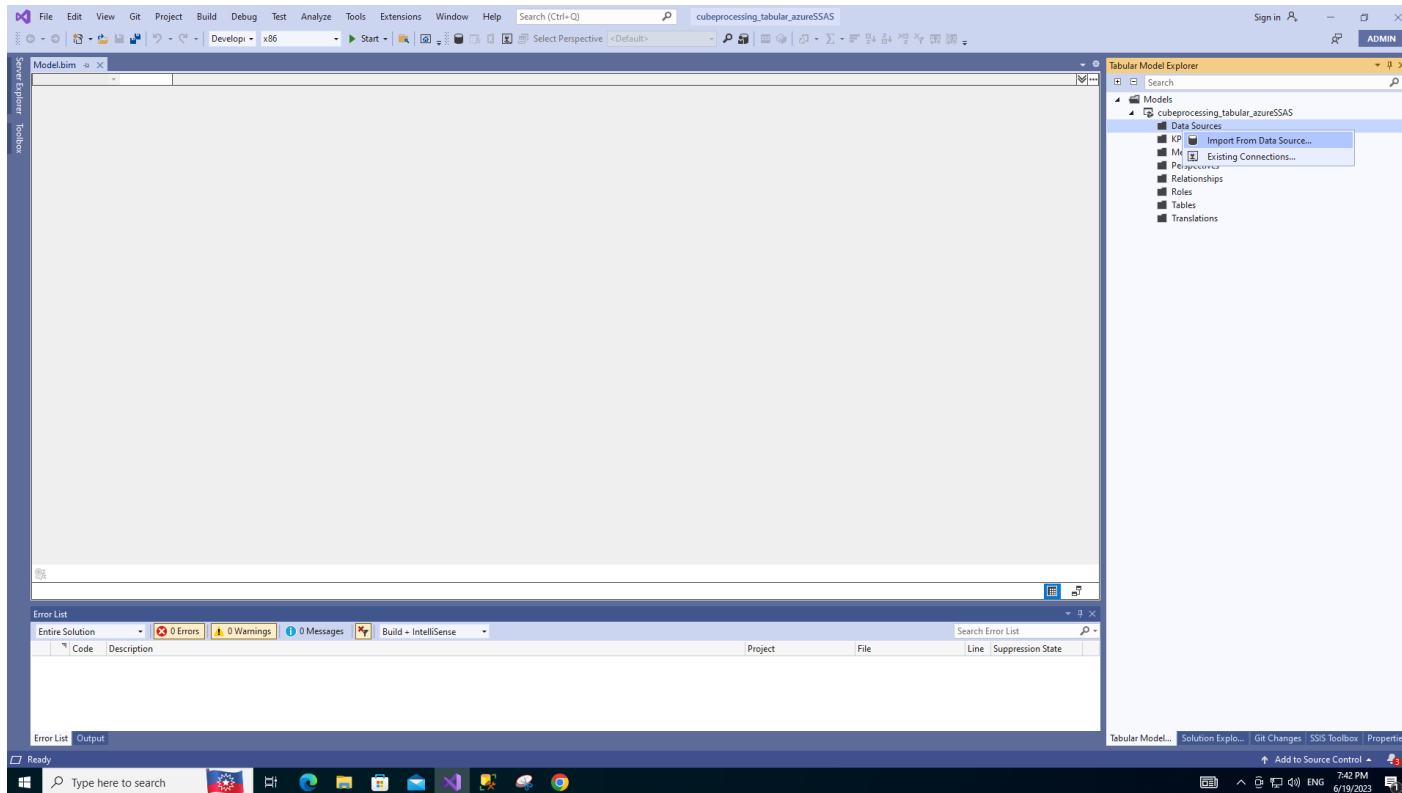
☒ Service Account
Connects to the data source using the credentials of the user running the Analysis Service server.

☐ Current User
Use the credentials of the current user.

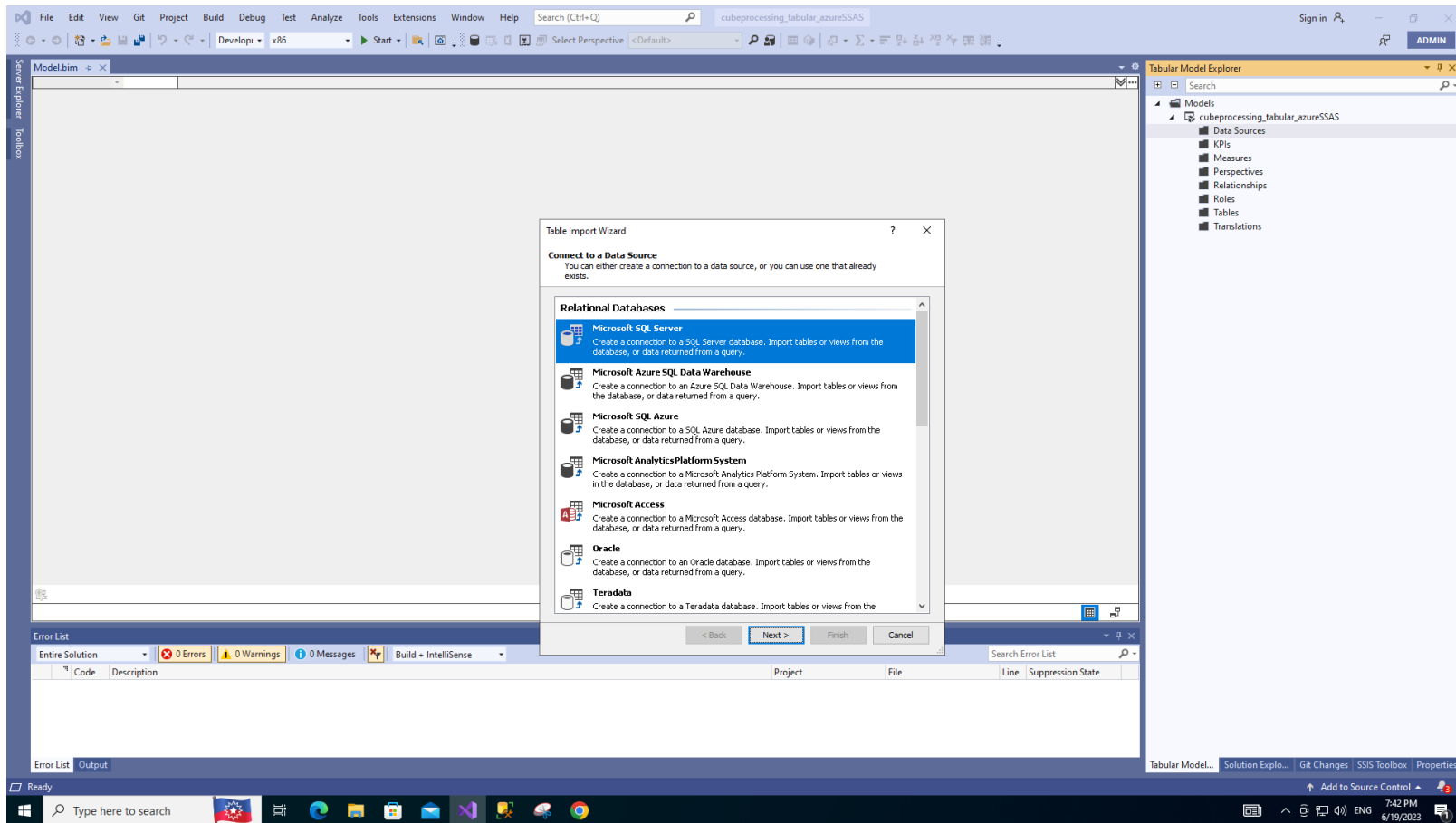
☐ Unattended Account
Connects to the data source using a low privilege account.

< Back Next > Finish Cancel

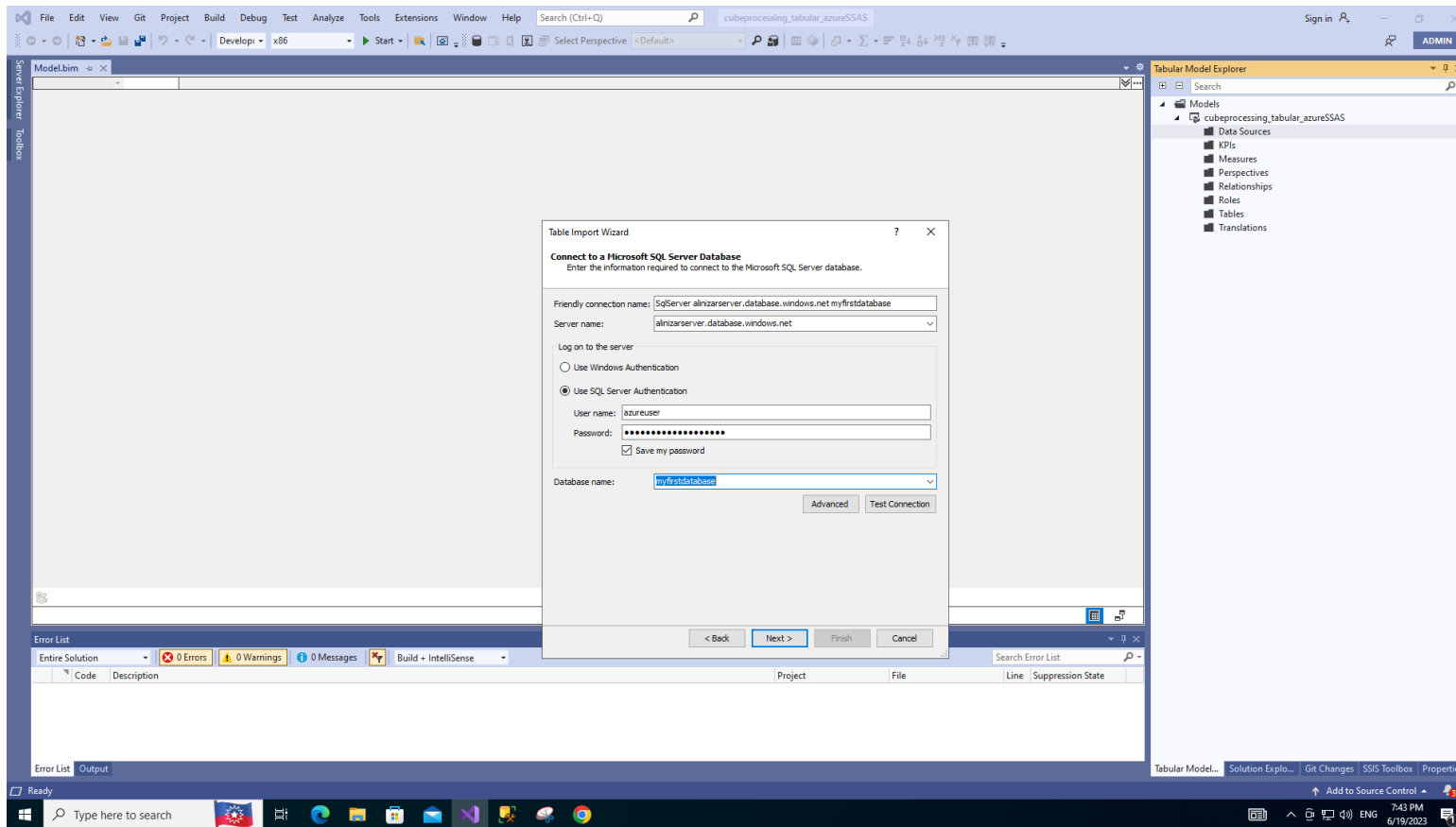
Step 4 – Import Data



Step 5 – Point to Azure SQL



Step 6 – Provide UserName/Password



Step 7 – Import Selected Data



The screenshot displays the SQL Server Enterprise Developer interface. The main window shows the 'Table Import Wizard' dialog box, which is used for importing data from a source table into a target table. The wizard is currently at the 'Select Tables and Views' step, where a list of tables and views is displayed. The 'vProductModelCatalogDescri...' table is selected. The 'Tabular Model Explorer' pane on the right shows the hierarchy of the tabular model, including Data Sources, KPIs, Measures, Perspectives, Relationships, Roles, Tables, and Translations. The 'vProductModelCatalogDescri...' table is highlighted under the 'Tables' folder.

Table Import Wizard

Select Tables and Views
Select the tables and views that you want to import data from.

Server: alnizarserver.database.windows.net
Database: myfirstdatabase

Tables and Views:

☐	Source Table	Schema	Friendly Name	Filter Details
☐	BuildVersion	dbo		
☐	ErrorLog	dbo		
☒	Address	Sales.T	Address	
☒	Customer	Sales.T	Customer	
☒	CustomerAddress	Sales.T	CustomerAddress	
☒	Product	Sales.T	Product	
☒	ProductCategory	Sales.T	ProductCategory	
☒	ProductDescription	Sales.T	ProductDescription	
☒	ProductModel	Sales.T	ProductModel	
☒	ProductModelProductDescrip...	Sales.T	ProductModelProductDescrip...	
☒	SalesOrderDetail	Sales.T	SalesOrderDetail	
☒	SalesOrderHeader	Sales.T	SalesOrderHeader	
☐	vGetAllCategories	Sales.T		
☐	vProductAndDescription	Sales.T		
☒	vProductModelCatalogDescri...	Sales.T		

Buttons: < Back, Next >, Finish, Cancel

Tabular Model Explorer

Models

- cubeprocessing_tabular_azureSSAS
 - Data Sources
 - KPIs
 - Measures
 - Perspectives
 - Relationships
 - Roles
 - Tables
 - Translations

Step 8 – Create measures



Visual Studio Code interface showing a data table, a context menu, and the Tabular Model Explorer.

Model.bim (Table: SalesOrderHeader)

idNumber	PurchaseOrderNumber	AccountNumber	Custo...	ShipToAddr...	BITToAddr...	ShipMethod	CreditCardApprovalCode	SubTotal	TaxAmt	Freight
1	PO348186287	10-4020-000609	29847	1092	1092	CARGO TRAN...		\$880.35	\$70.43	\$22.0
2	PO19952192051	10-4020-000106	30072	640	640	CARGO TRAN...		\$78.81	\$6.30	\$1.9
3	PO19604173239	10-4020-000340	30113	653	653	CARGO TRAN...		\$38,418.69	\$3,073.50	\$960.4
4	PO19372114749	10-4020-000582	29485	1086	1086	CARGO TRAN...		\$39,785.33	\$3,182.83	\$994.63
5	PO19343113609	10-4020-000024	29957	992	992	CARGO TRAN...		\$83,858.43	\$6,708.67	\$2,096.46
6	PO19285135919	10-4020-000448	29736	659	659	CARGO TRAN...		\$108,561.83	\$8,684.95	\$2,714.05
7	PO17052159664	10-4020-000420	29660	1058	1058	CARGO TRAN...		\$57,634.63	\$4,610.77	\$1,440.87
8	PO16501134889	10-4020-000140	29796	642	642	CARGO TRAN...		\$78,029.69	\$6,242.38	\$1,950.74
9	PO13021155785	10-4020-000276	30089	1034	1034	CARGO TRAN...		\$1,141.58	\$91.33	\$38.54
10	PO12992180445	10-4020-000295	30027	1038	1038	CARGO TRAN...		\$3,398.17	\$271.85	\$84.95
11	PO10295111084	10-4020-000322	30019	652	652	CARGO TRAN...		\$2,016.34	\$161.31	\$50.41
12	PO10353140756	10-4020-000088	29922	639	639	CARGO TRAN...		\$35,775.21	\$2,862.02	\$894.38
13	PO2697119362	10-4020-000187	29938	1020	1020	CARGO TRAN...		\$41,622.05	\$3,329.76	\$1,040.55
14	PO2378131604	10-4020-000635	30102	669	669	CARGO TRAN...		\$2,453.76	\$196.30	\$61.34
15	PO16530177647	10-4020-000601	30033	1090	1090	CARGO TRAN...		\$602.19	\$48.18	\$15.05
16	PO16153112278	10-4020-000186	29653	1019	1019	CARGO TRAN...		\$13,823.71	\$1,105.90	\$345.99
17	PO16124166561	10-4020-000649	29975	1098	1098	CARGO TRAN...		\$3,324.28	\$265.94	\$83.11
18	PO13050111529	10-4020-000160	29644	643	643	CARGO TRAN...		\$1,059.31	\$84.74	\$26.48
19	PO6119130779	10-4020-000268	29612	649	649	CARGO TRAN...		\$550.39	\$44.03	\$13.76
20	PO3770176273	10-4020-000151	29584	1014	1014	CARGO TRAN...		\$246.74	\$19.74	\$6.17
21	PO4785152479	10-4020-000223	29877	1026	1026	CARGO TRAN...		\$12,685.89	\$1,014.87	\$317.15
22	PO5713190501	10-4020-000052	29932	637	637	CARGO TRAN...		\$63,980.99	\$5,118.48	\$1,599.52
23	PO4582142611	10-4020-000025	29568	993	993	CARGO TRAN...		\$2,415.67	\$193.25	\$60.39
24	PO5529125166	10-4020-000061	29920	999	999	CARGO TRAN...		\$74,058.81	\$5,924.70	\$1,851.47
25	PO2349143275	10-4020-000006	29638	989	989	CARGO TRAN...		\$2,137.23	\$170.98	\$53.43
26	PO29111718	10-4020-000304	30025	651	651	CARGO TRAN...		\$40.90	\$3.27	\$1.02

Context Menu: Sum, Average, Count, DistinctCount, Max, Min

Tabular Model Explorer:

- Models
 - cubeprocessing_tabular_azureSSAS
 - Data Sources
 - SqlServer analyzer.database.windows.net myfirstdata
 - Measures
 - Total_LineTotal
 - Perspectives
 - Relationships
 - CustomerAddress [AddressID] - Address [AddressID]
 - CustomerAddress [CustomerID] - Customer [CustomerID]
 - Product [ProductCategoryID] - ProductCategory [ProductCategoryID]
 - Product [ProductModelID] - ProductModel [ProductModelID]
 - ProductModelProductDescription [ProductDescriptionID] - ProductModelProductDescription [ProductDescriptionID]
 - SalesOrderDetail [ProductID] - Product [ProductID]
 - SalesOrderDetail [SalesOrderID] - SalesOrderHeader [SalesOrderID]
 - SalesOrderHeader [BillToAddressID] - Address [AddressID]
 - SalesOrderHeader [CustomerID] - Customer [CustomerID]
 - SalesOrderHeader [ShipToAddressID] - Address [AddressID]
 - Roles
 - Tables
 - Address
 - Customer
 - CustomerAddress
 - Product
 - ProductCategory
 - ProductDescription
 - ProductModel
 - ProductModelProductDescription
 - SalesOrderDetail
 - Columns
 - Hierarchies
 - SalesOrderHeader
 - Columns
 - Hierarchies
 - Measures
 - Total_LineTotal
 - Partitions

Step 9 – deploy to server



A screenshot of the Visual Studio Code interface. The main editor window displays a table with columns: Id, Number, PurchaseOrderNumber, AccountNumber, Custid, ShipToAddress, BillToAddress, ShipMethod, CreditCardApprovalCode, SubTotal, Taxamt, Freight, TotalDue, Comment, rowguid, and ModifiedDate. The table contains 26 rows of data. The Solution Explorer on the right shows a project named 'cubeprocessing.tabular.azureSSAS'. The Error List at the bottom shows a warning: 'Using the service account as the impersonation setting for a data source is less secure than using a Windows account. Select Existing Connections from the Model menu, select a data source, and then click Edit to change the impersonation settings.' The status bar at the bottom indicates 'Ready' and 'Type here to search'.

Tutorial



Students can also the following example from Microsoft for Furter reading

<https://docs.microsoft.com/en-us/analysis-services/tutorial-tabular-1400/as-lesson-1-create-a-new-tabular-model-project?view=asallproducts-allversions>